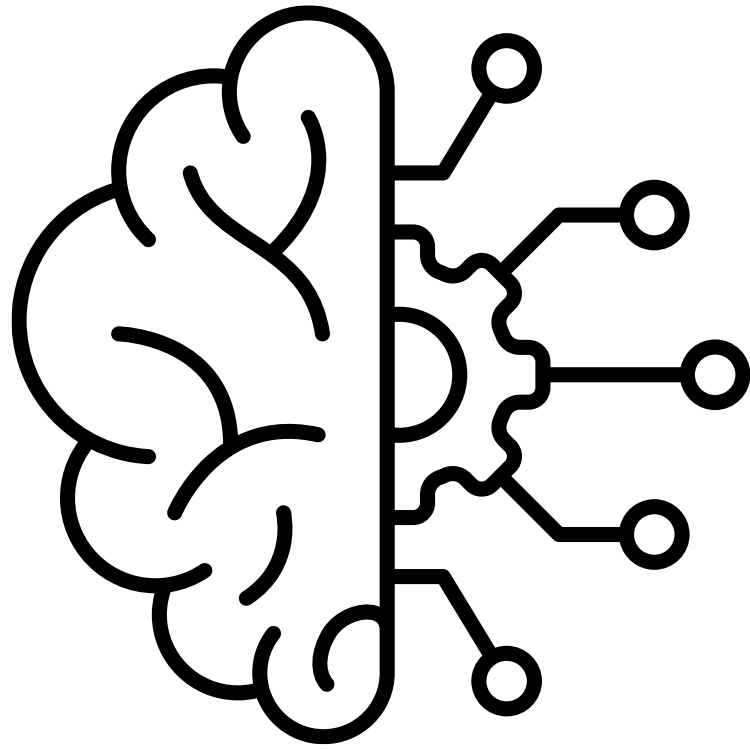




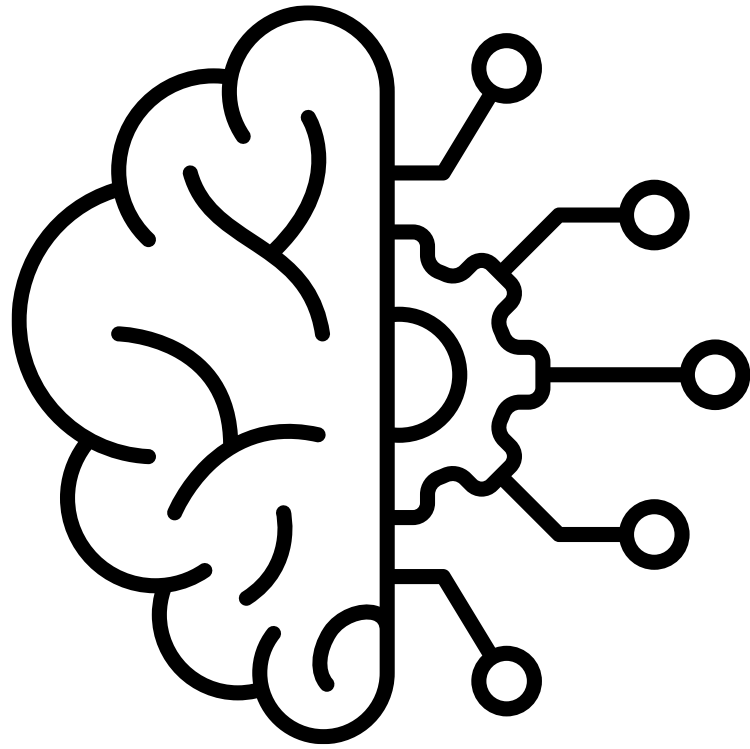
Building AI Agents for the Humanities

**Session 1b: Foundations of AI Agents &
Agentic AI**

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Postdoctoral Fellow, Western University
Assistant Professor of Digital Humanities





Building AI Agents for the Humanities

- **Foundations of AI Agents & Agentic AI**
- **AI Agents Components**
- **N8N Quick start**

~80 years of AI

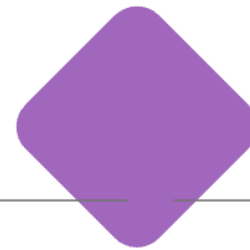
1950s-1960s



Foundations

- Turing Test (1950) – A Benchmark for machine intelligence.
- Dartmouth Conference (1956) – Formalized Artificial Intelligence as a research field.
- ELIZA (1966) – Pioneered pattern-matching for conversational AI.

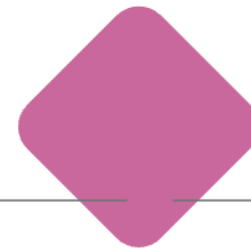
1970s - 1980s



Rule-Based Systems

- Expert systems like MYCIN applied rule-based logic to medical diagnosis.
- PROLOG brought logic programming into AI development.
- Core principles of reinforcement learning were established.

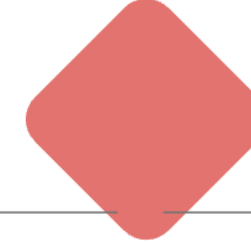
1990s



Intelligent Agents Emerge

- Systems began autonomous information gathering and processing
- Early virtual assistants appeared in consumer and enterprise settings.

2000s



Machine Learning Dev

- Advance Natural Language Processing (IBM Watson)

2010s



DL Revolution

- AlexNet (2012) transformed image recognition through deep neural networks.
- 2017 “Attention is all that matters.”
- GPT-3 (2020) achieved near-human-level text generation.

2020s



Agentic AI

- RAG Systems (context & memory capabilities)
- Multi-Agent Systems
- Agentic Architectures

Types of Agents

Symple Reflex

Simple reflex agents – Act only on the current situation using predefined rules, without retaining memory or learning.

Utility Based Agent

Evaluate different outcomes with a utility function and select the action that maximizes overall benefit.

Goal-based agents

Choose actions by considering how they contribute to achieving specific goals, enabling better planning and flexibility.

Model-based agents

Use an internal model of the world to handle partially observable situations and make more informed decisions.

Learning agents

Improve their performance over time by learning from past experiences and adapting their behaviour.

Hierarchical agents

Organize decision-making into multiple levels (low-level reflexes, mid-level planning, high-level strategy) to manage complex tasks.

Agent Taxonomy

Core Functional Capabilities

Memory Agents

- Short-Term Memory Agent
- Long-Term Memory Agent
- Selective Forgetting Agent



Processing and Reasoning

- Narrow AI Agent
- General AI Agent



Perception and Input Modality

- Single Modality Agent
- Multi-Modality Agent



Action and Output Execution

- Tool Agent
- Embodied Agent



Role and Scope

Objective and Goal Orientation

- Reactive Agent
- Proactive Agent



Task Specialization

- Task-Specific Agent
- General-Purpose Agent



Structural Dimensions

Coordination and Communication

- Multi-Agent
- Single Agent
- Competitive Agent
- Collaborative Agent



Planning Mechanism

- Internal Planning Agent
- External Planning Agent



System Architecture

- Homogenous Agent
- Heterogenous Agent



Scalability and Deployment

- Local Deployment Agent
- Distributed System Agent



Non-Functional Qualities

Efficiency and Performance

- High-Speed Agent
- High-Accuracy Agent



Ethics and Trust

- Explainable Agent
- Fairness-Oriented Agent

Robustness and Resilience

- Fault-Tolerant Agent
- Dynamic Environmental Agent



Interaction with System

Integration of External Capabilities

- API-Driven Agent
- UI-Driven Agent



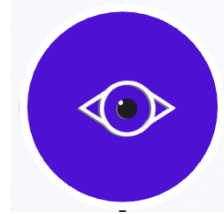
Environment and Context Interactions

- Virtual Agent
- Physical Agent



Again... what is an AI Agent?

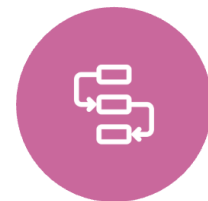
An **AI agent** is a system that:



Perceives its environment (input/data/trigger)



Decides what to do (reasoning/planning)



Acts to achieve a goal (takes actions, often over time)



Adapts based on feedback (optional but common)

Travel Planning Agent

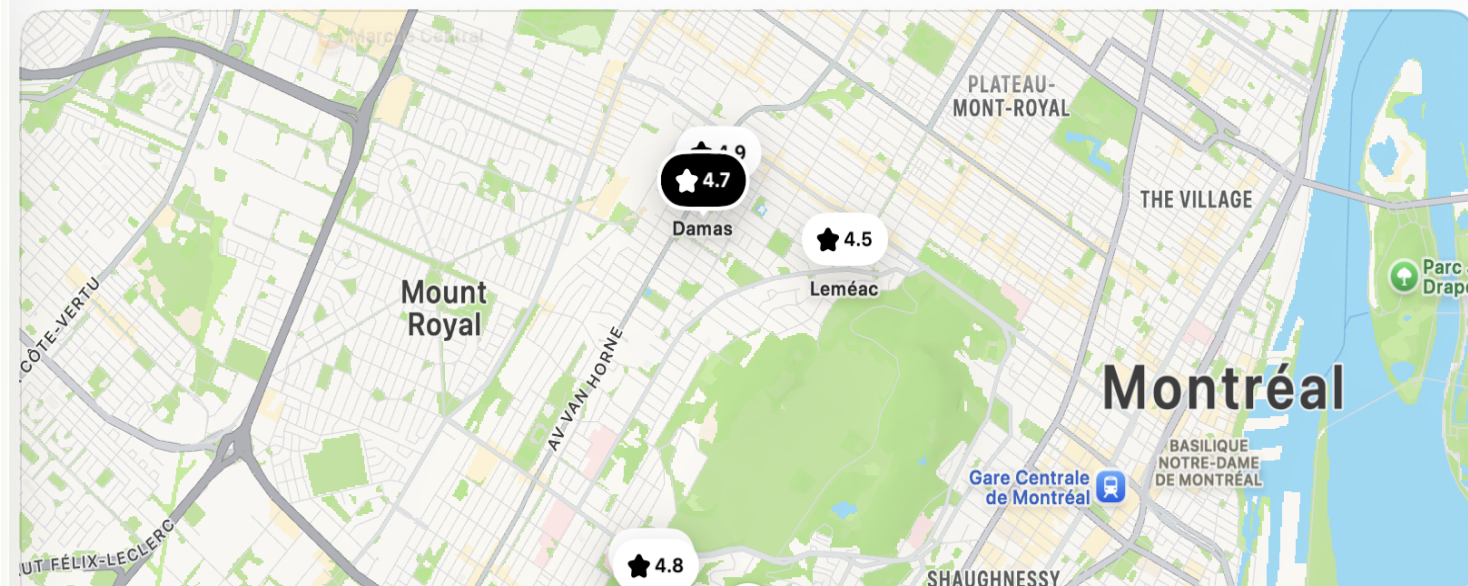
Text Input/Output

Here's a concise dinner plan near Université de Montréal / Outremont area (all well-reviewed and close by):

- **June 8 — Damas**
Upscale Syrian cuisine; one of the top-rated restaurants in the area.
- **June 9 — Leméac**
Classic French bistro; reliable and elegant neighborhood favorite.
- **June 10 — Restaurant Ermitage : Authentic European Cuisine and Fine Dining**
Cozy Eastern European spot; highly rated and close to campus.
- **June 11 — Myers**
Small, contemporary restaurant; creative menu and excellent reviews.
- **June 12 — Chez Lee**
Casual, local favorite; great for a relaxed, quick dinner.

These are all within walking distance or a short transit ride from Université de Montréal, and consistently well reviewed.

tripadvisor.ca

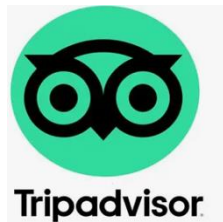


Goal: Schedule my dinners in Montreal from June 8th to June 12th at restaurants near the University of Montreal with good reviews after my last activity of the day.

Ask Google Maps API for all restaurants closer to the Univ of Montreal



Filter those with good reviews from TripAdvisor

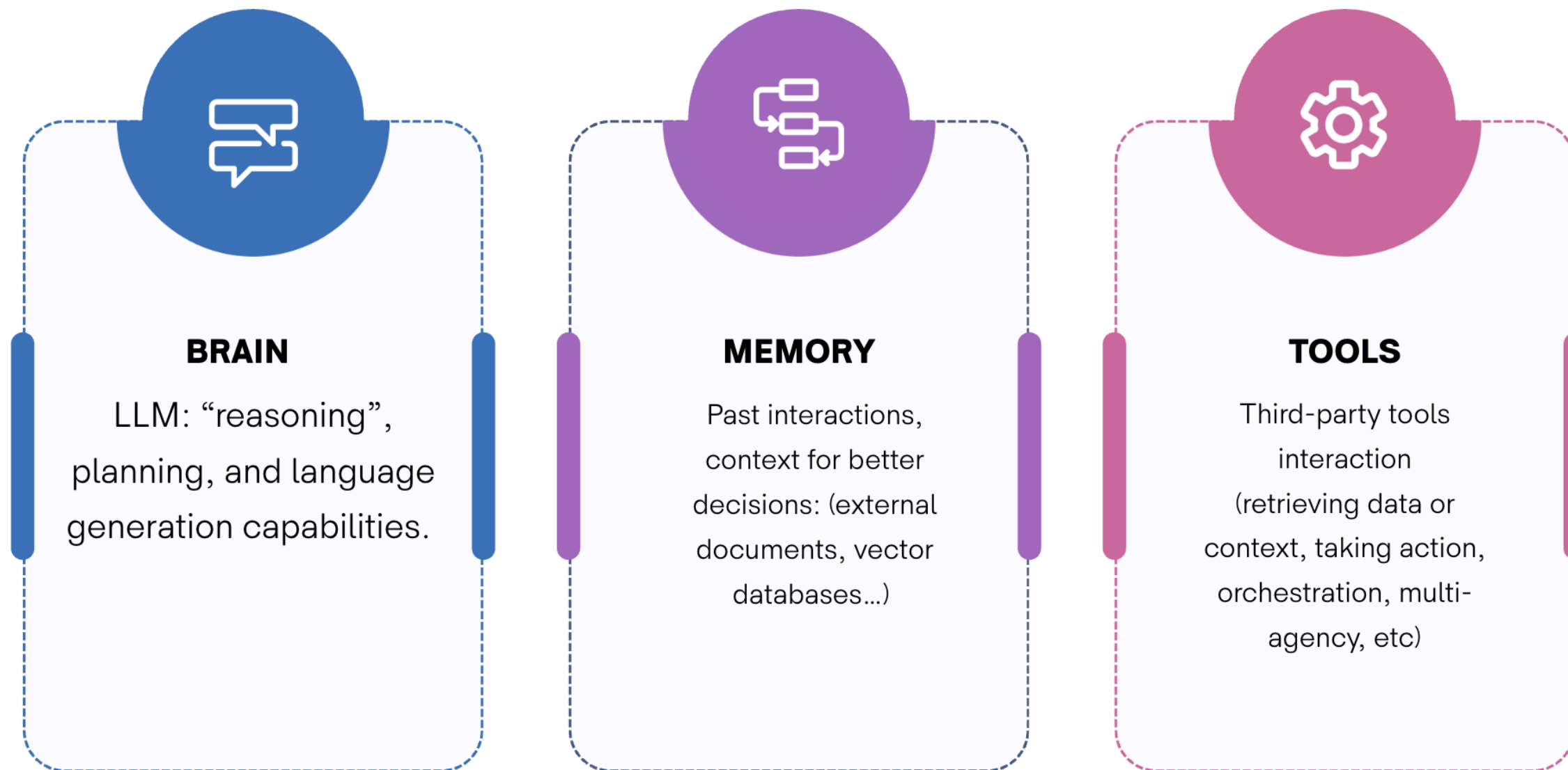


Schedule the dates and times based on the availability on your calendar

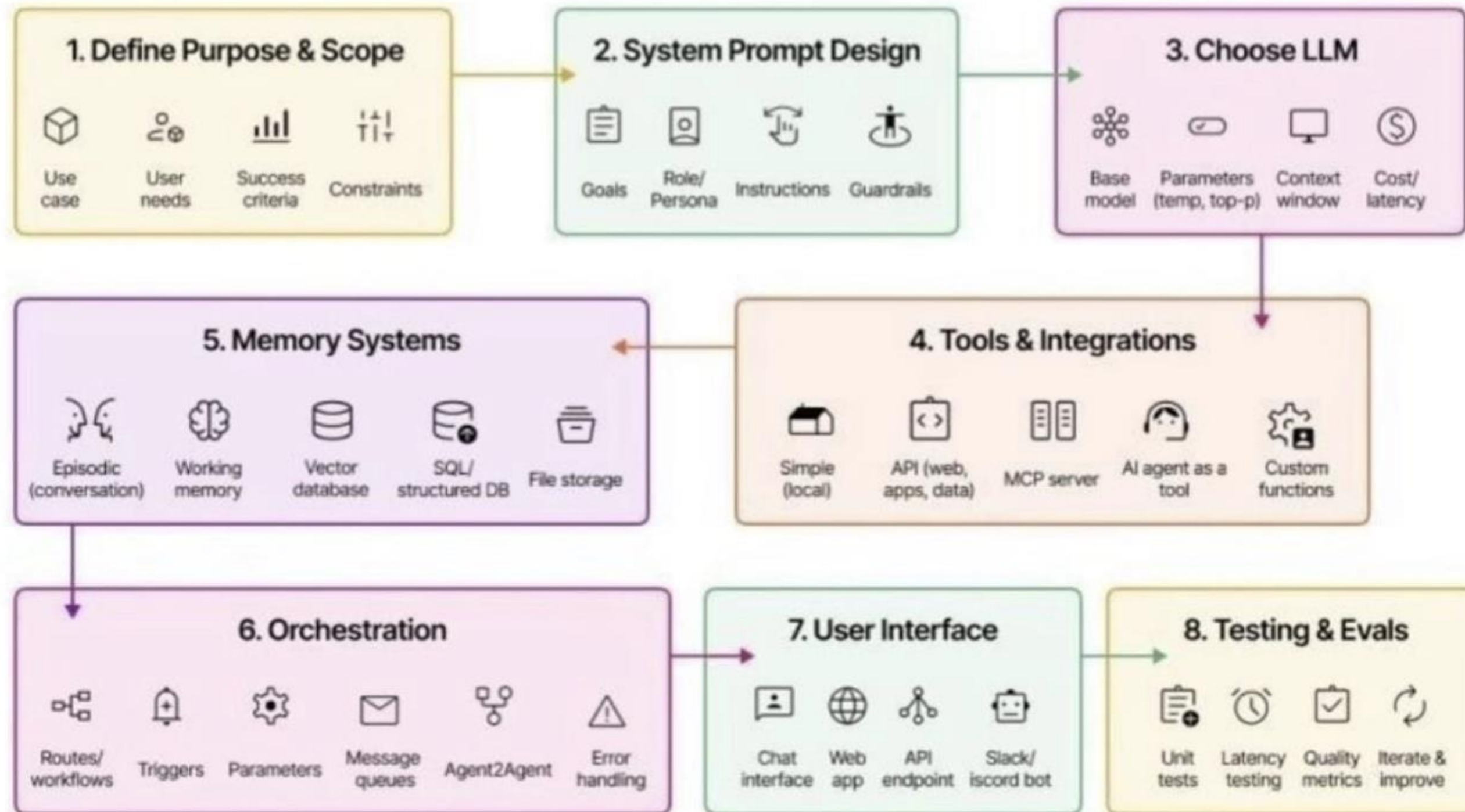


Book the reservation (Limitations?)

Key Components of an AI Agent



The Process of Building an AI Agent



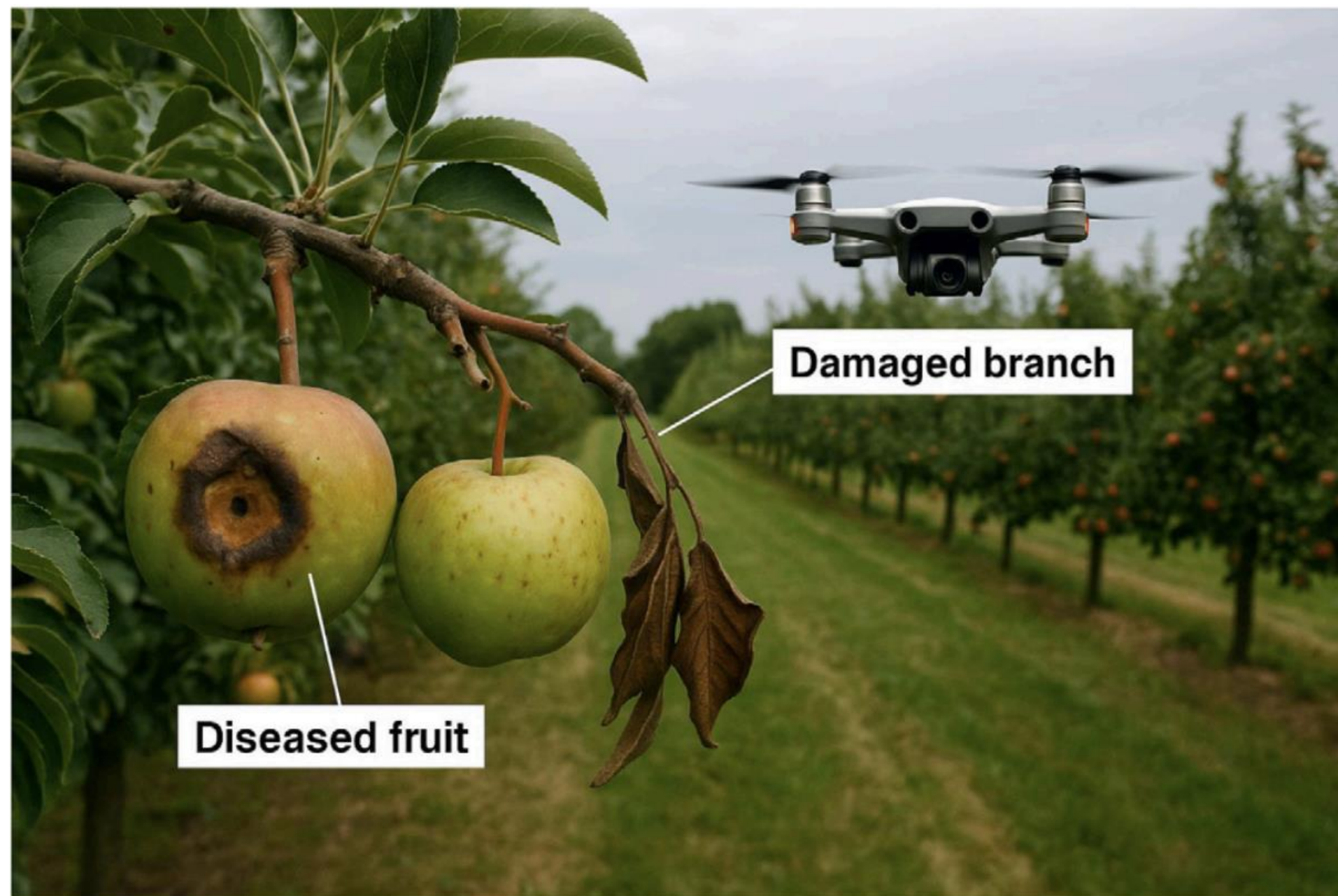
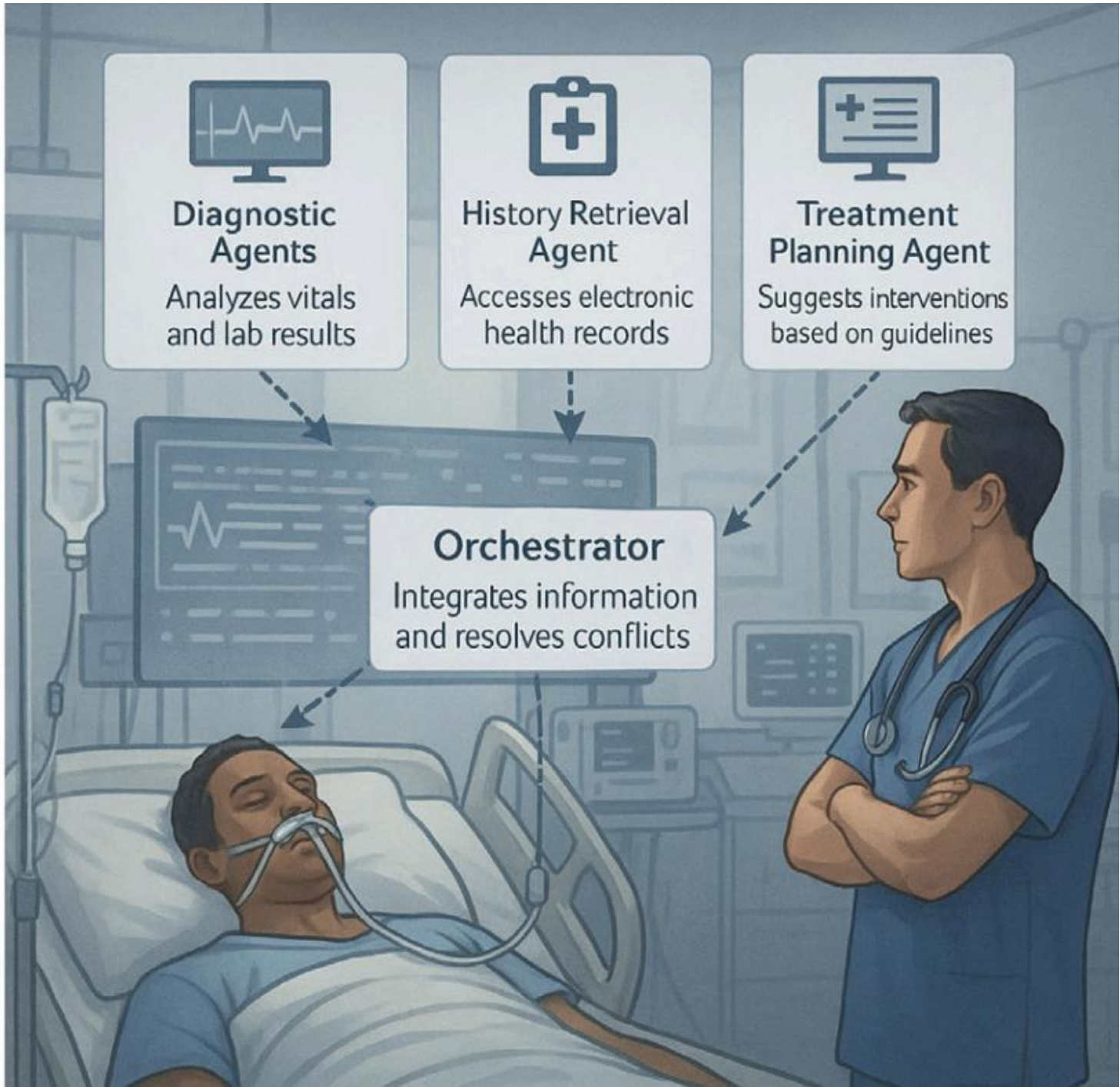


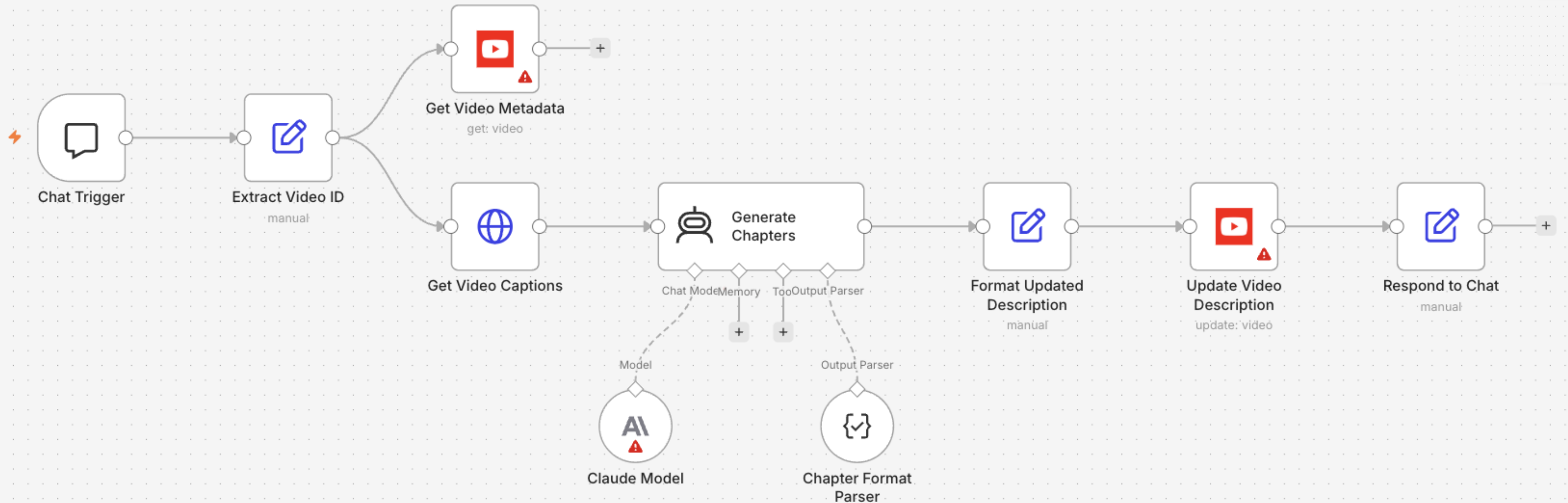
Fig. 5. An AI agent-enabled drone autonomously inspects an orchard, identifying diseased fruits and damaged branches using vision models, and triggers real-time alerts for targeted horticultural interventions.



Clinical decision support in hospital ICUs through synchronized agents for diagnostics, treatment planning, and EHR analysis, enhancing safety and workflow efficiency.

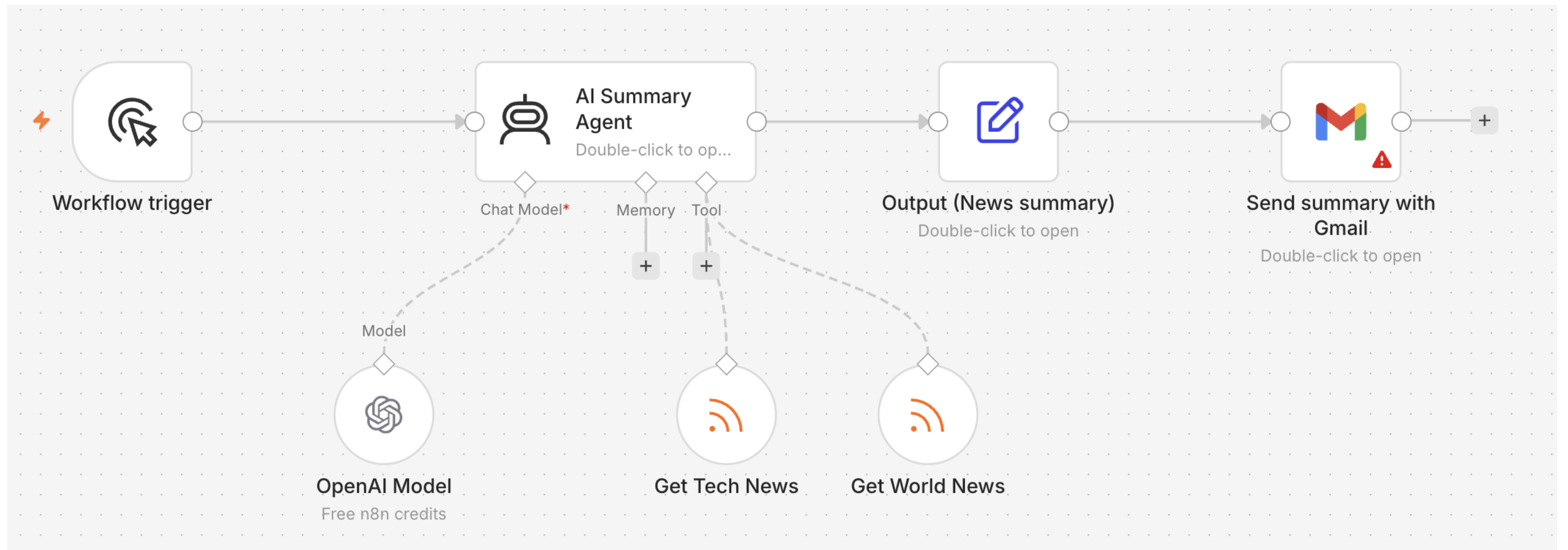
Video Editing

Goal: Summarize



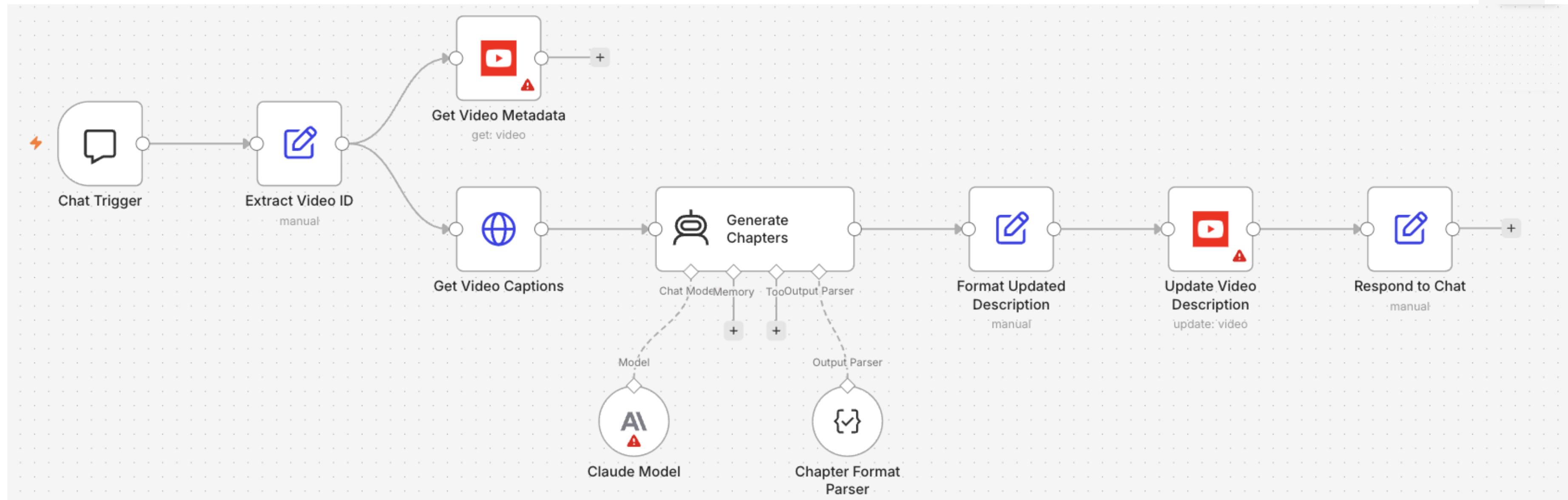
News Intelligence Agent (Basic)

Goal: Provide a summary of world and tech news from the last 24 hours.



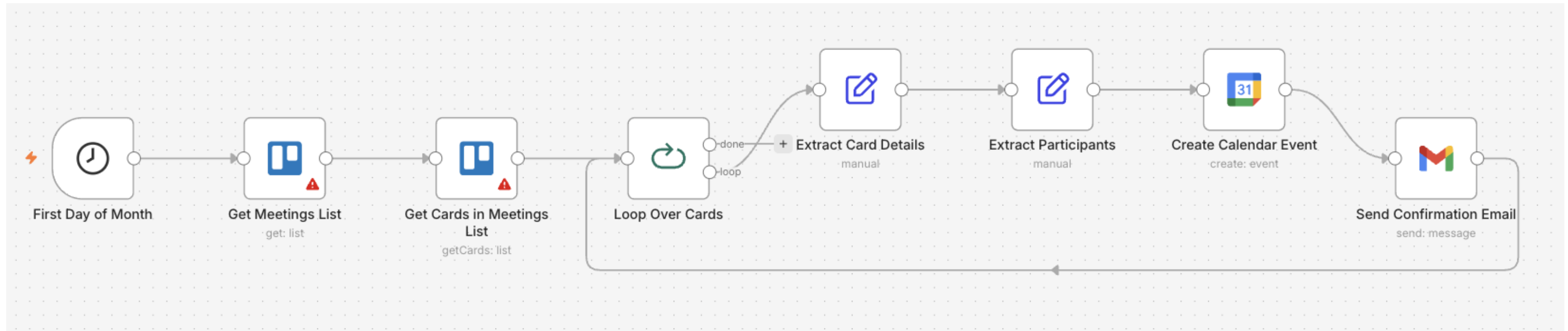
Video Editing

Goal: Generate YouTube chapter timestamps from Video Captions



Project Manager Meetings

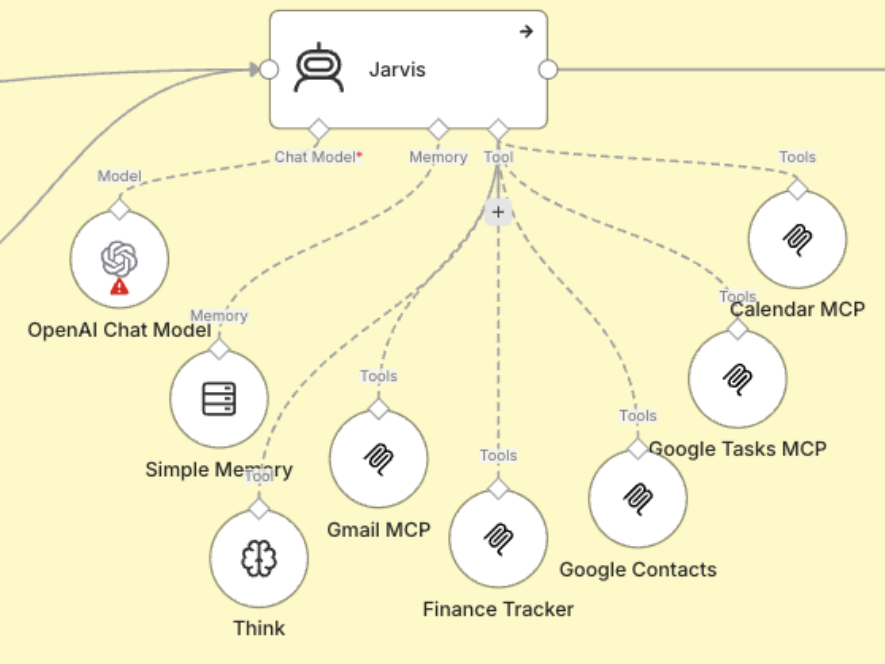
Goal: Generate YouTube chapter timestamps from Video Captions



Jarvis 🤖

Your AI-powered personal assistant.

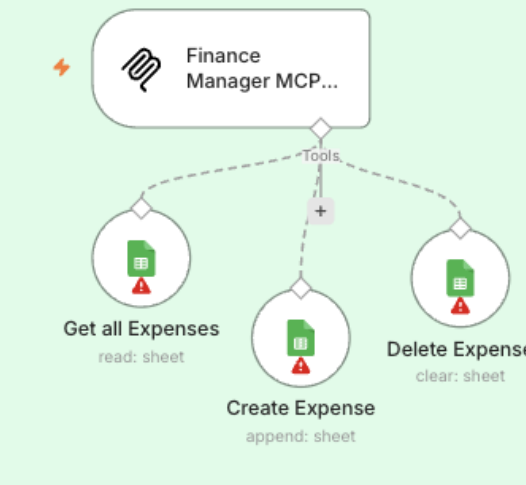
- Orchestrates tasks, calendar, emails, contacts & expenses
- Uses memory + OpenAI model for smart decisions
- Sends results back to Telegram



Finance Manager MCP 📊

Track personal or business expenses.

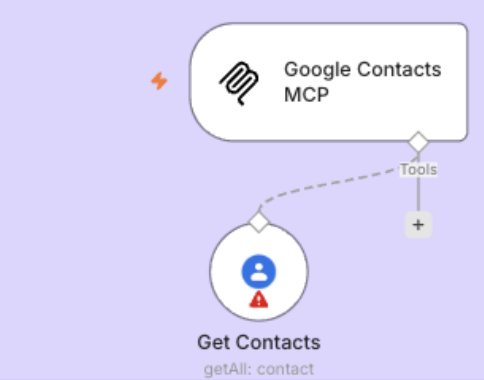
- Create new expenses
- Get expense reports
- Delete outdated entries



Google Contacts MCP 👤

Access and manage your contact list.

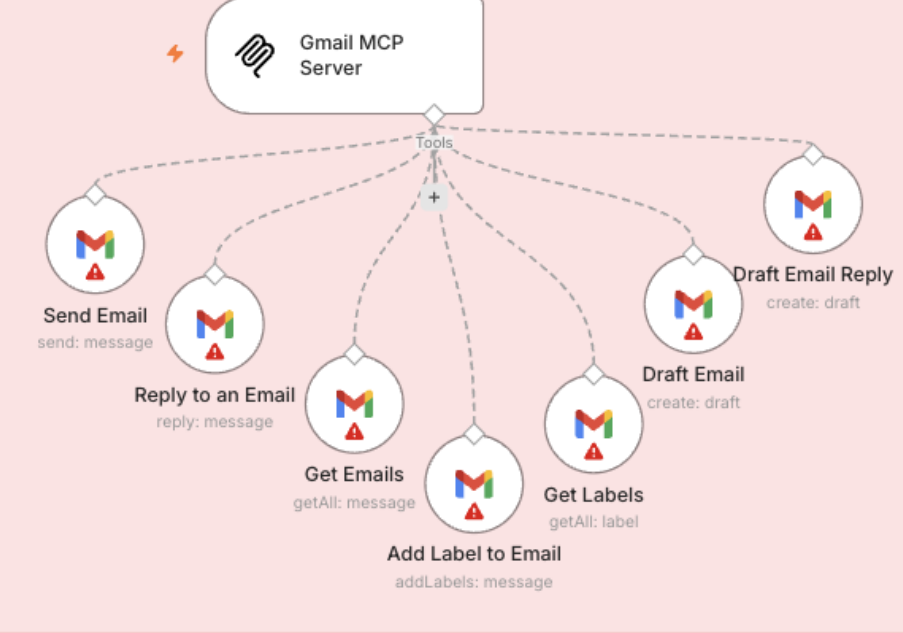
- Fetch contacts for quick communication



Gmail MCP 📧

Full email management.

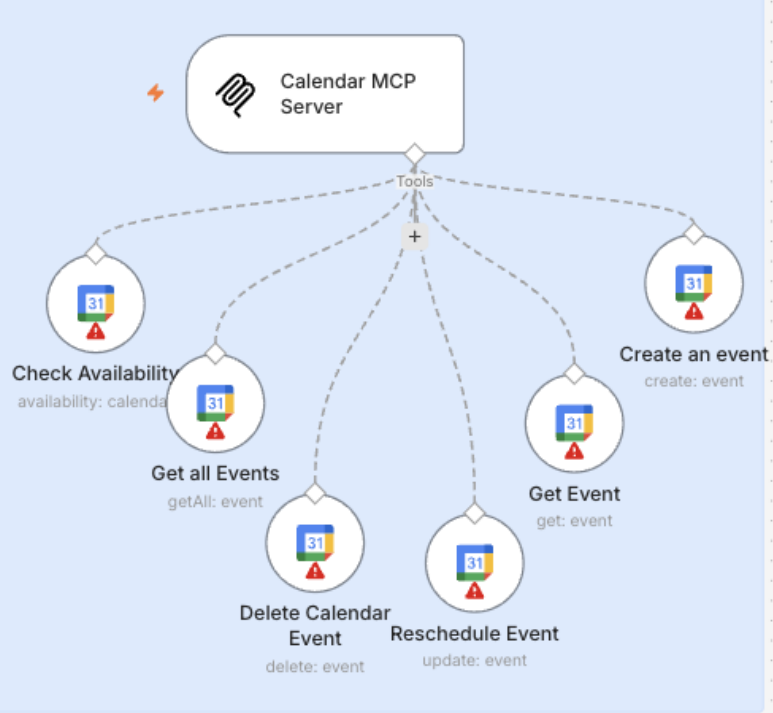
- Send & draft messages
- Reply, label, and fetch emails



Calendar MCP 📅

Your scheduling hub.

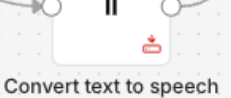
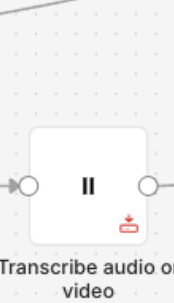
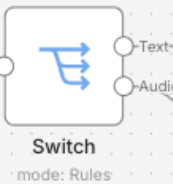
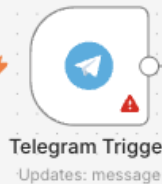
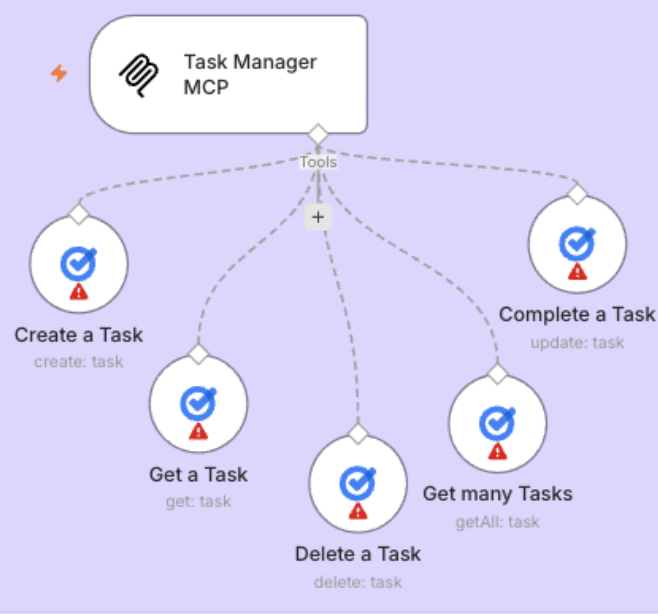
- Check availability
- Create, reschedule, or delete events

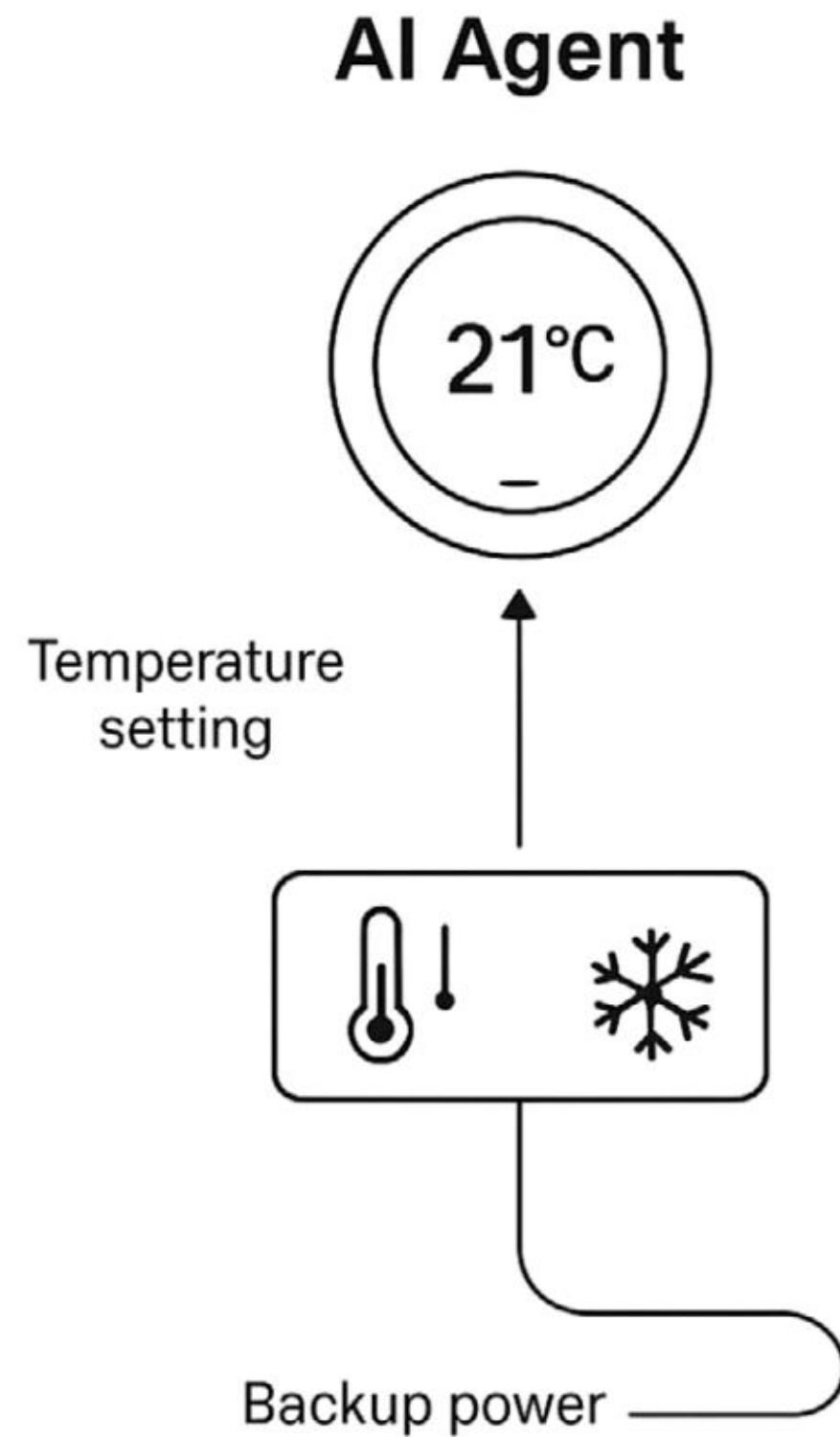


Task Manager MCP ✅

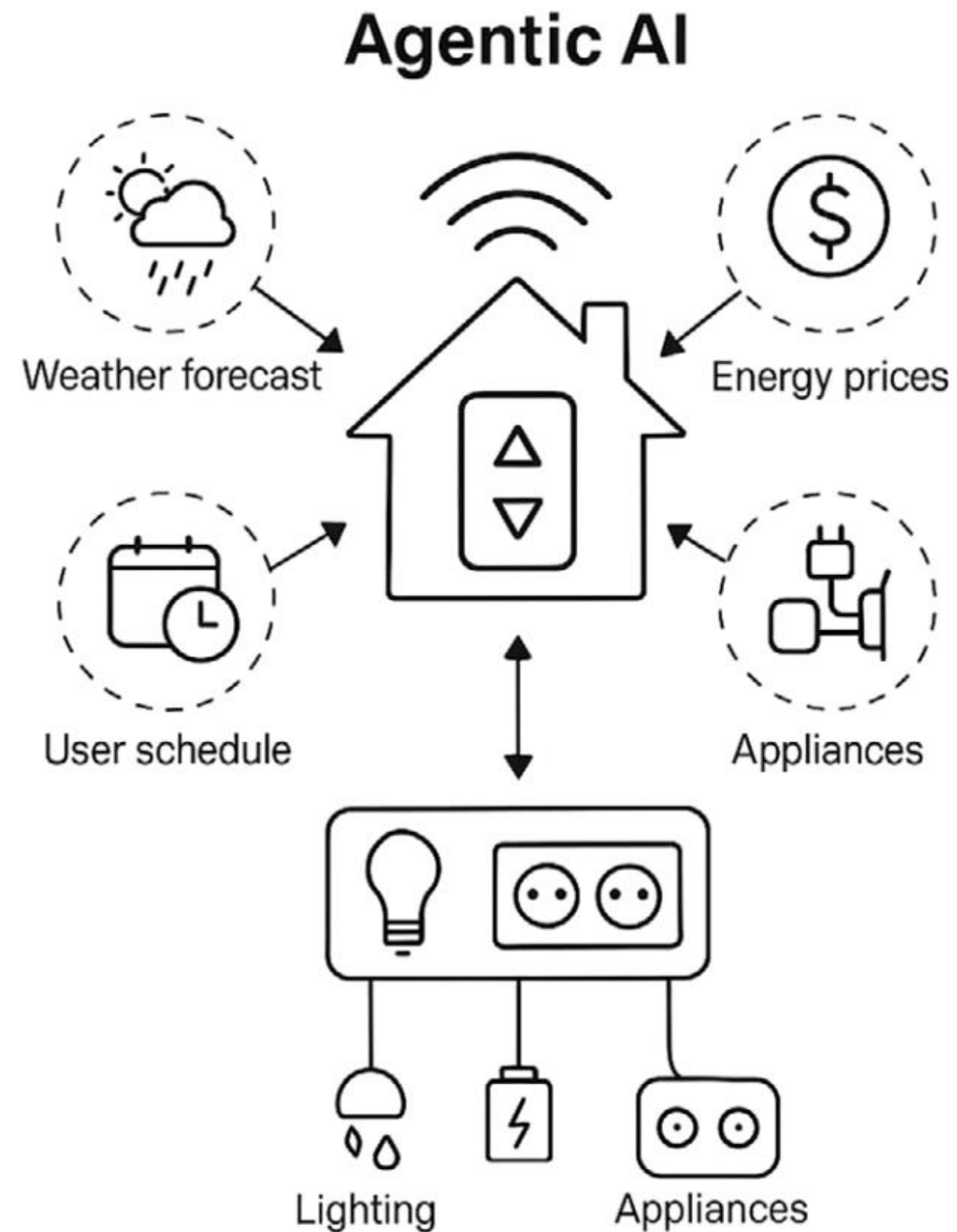
Manages to-dos with ease:

- Create / Complete / Delete tasks
- Retrieve individual or bulk tasks





Basic AI Agent managing a single task — such as maintaining a thermostat's temperature setting based on user input.



Agentic AI system coordinating multiple agents to manage household automation tasks including temperature regulation, weather forecasting, energy pricing, appliance scheduling, and security monitoring.



Let's Practice

Design Challenge

In groups, design an AI agent that can achieve a specific goal.

Your agent must include:

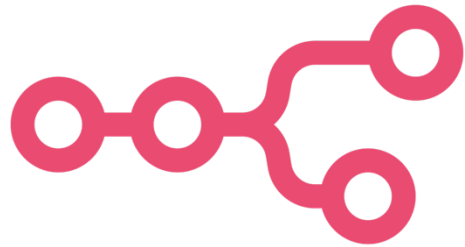
- **Goal:** What is the agent trying to accomplish?
- **Prompt(s):** What instructions guide the agent?
- **Steps:** What is the sequence of actions or decisions?
- **Tools:** What tools or data sources does it use (e.g., search, APIs, documents)?
- **Human Role:** Where does human input or judgment matter?

Be prepared to present:

- What does your agent do? (Goal)
- How does it work (process + logic)?
- Why is it useful?
- What are its limitations?



Installation Guide and Quick Start



n8n

Canvas Overview

 n8n + 🔍 📄

🏠 Overview

💬 Chat Preview

📦 Templates

📊 Insights

🔗 Help

⚙️ Settings

👤 Personal / My workflow + Add tag

0 / 1

Publish

▼



Editor

Executions

Evaluations



Add first step...

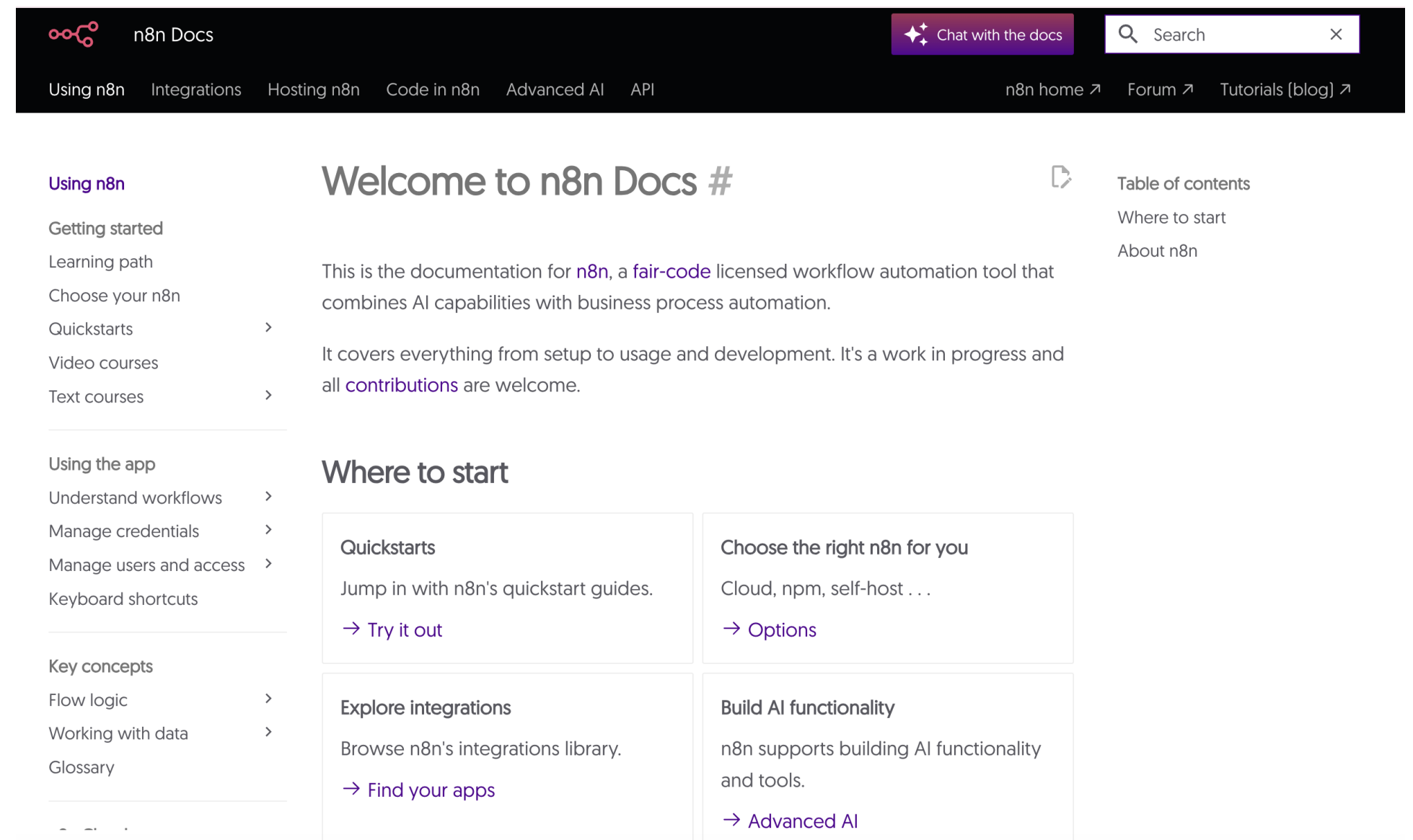
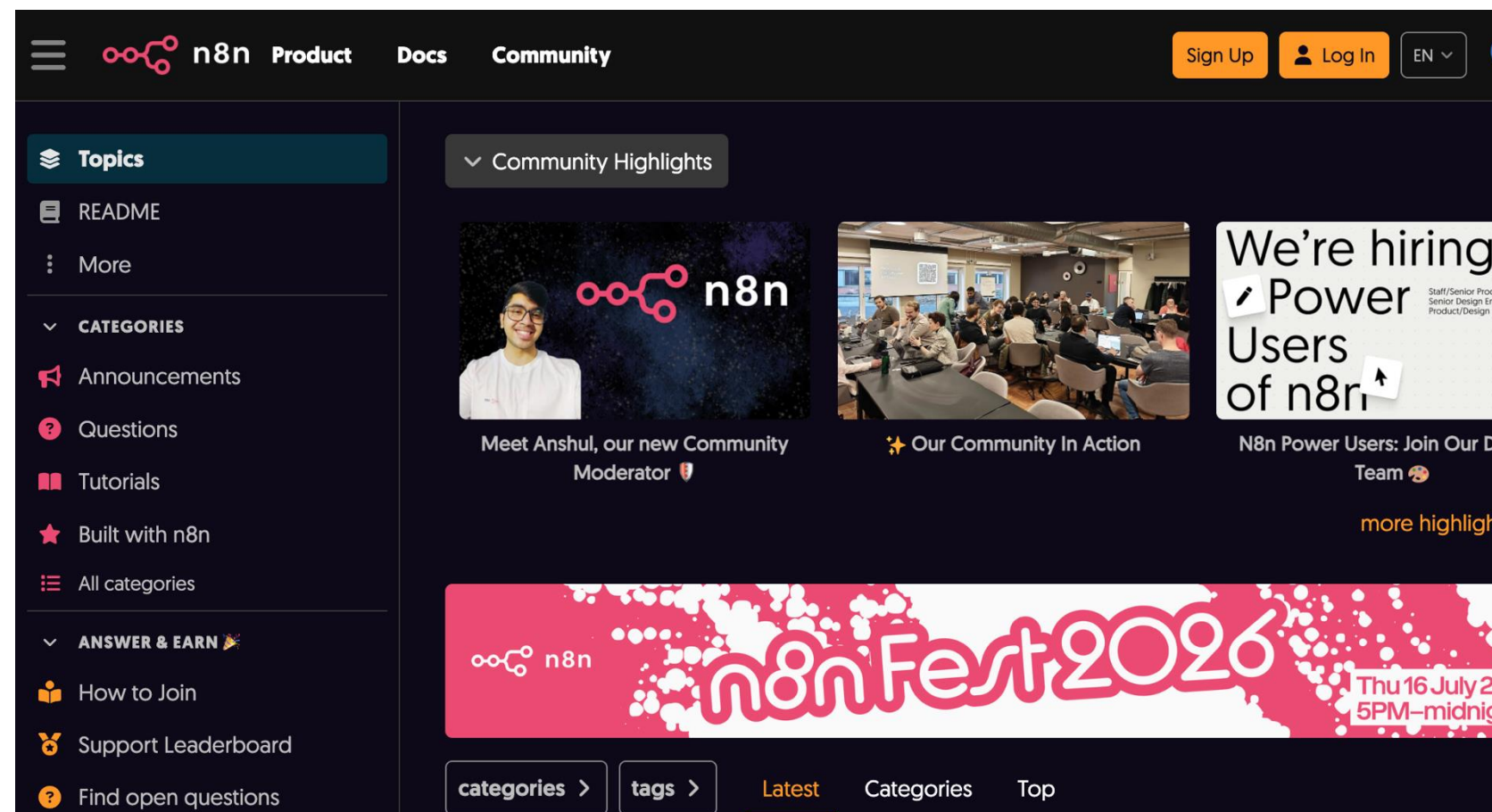


Logs



Extending your practice

- Documentation: <https://docs.n8n.io/>
- Templates: <https://n8n.io/workflows/>
- Community: <https://community.n8n.io/>



Other Alternatives

- AgentX: AI Native, multi-agent automation
- Zapier: Fully conversational
- Pipedream: Code-friendly, Serverless function
- Huginn: Self-Hosted, Open Source
- Openclaw: Free, Open Source, Local
- Claude Agents
- and many more....

Challenges & Considerations

- Costs (Token-Based Business Models)
- Integration complexity (technical skills)
- Trust and Control Mechanisms
- Human-AI collaboration > Full Agent Autonomy
- Sustainability (Do I really need an agent for this?)
- Cognitive Skills Development

Skills development

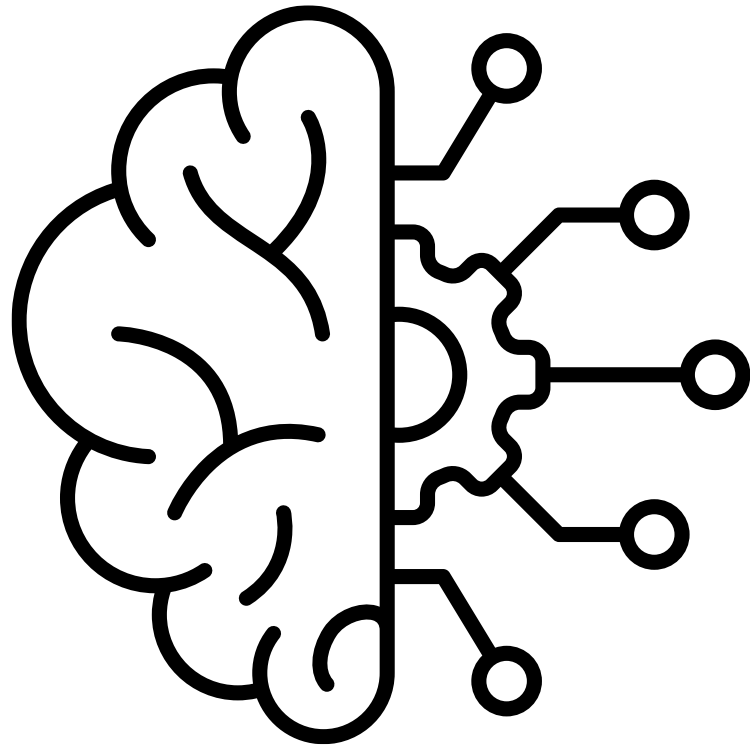
Humans can provide:

- Creative thinking
- Context understanding
- Ethical judgment
- Emotional intelligence
- Interpersonal trust-building
- Complex decision- making

Agents can provide:

- Information processing
- Pattern recognition
- Consistent execution
- Tireless operation
- Scalable application of learned knowledge





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